Implementation of Smart Home through FPGA using Verilog Hardware Descriptive Language

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***Abstract*— Home Automation involves intelligent control of the electrical and electronic devices of a home, office or building. A project of a home automation system is made in order to control two integral parameters security and comfort of a home. These include lighting and temperature control for comfort. The security system detects the fire and intruders through the door, window and garage. Digital systems design is the foundation of smart home project. Comfort and security of a house are controlled with the help of this project. Smart home is implemented through a smart verilog code in this paper. Security and comfort is controlled by interfacing devices and sensors to Field programmable gate array abbreviated as FPGA. The verilog code has been synthesised on Xilinx platform. FSM concepts are used. Simulated waveforms are obtained and verified.**

Keywords*— FPGA, Xilinx, Moore Machine, Home Automation System, State diagram, Simulation, Synthesis, verilog HDL

I. INTRODUCTION

Now days everyone wants to be at ease when it comes to home. Security is the prerequisite for everyone. Such needs can be accomplished by a Smart Home. The security module is responsible for providing the security against thief and mishaps. The comfort module provides comfortable surroundings for the owner of the home. Security is of great importance whereas comfort is bliss when a home is taken into account. These two modules are integrated in order to modernize a home. The home automation system keep owner secure and make the home as comfortable as possible. This paper consists of a verilog code that activates security and then comfort module. This allows the owner to relax. He need not worry to check doors repeatedly. This project is executed with two steps. The moment owner arrives at home, he enters a password to get inside the home and home automation module gets activated. Security module gets activated as soon as right password is entered. Comfort Module is activated as soon as motion sensor signal is high. When the owner is not at home only the password module gets triggered. Various sensors are interfaced in this project. These sensors behave as input to the program. Xilinx Spartan 3E FPGA is used in this project.

A. Ambition

Smart home project is made to include security aspect alongwith comfort module for the owner of the house with lowest possible cost and maximum features.

B. Aim

The aim of this project is to develop a smart home that encapsulates security and comfort features for owner of the home.

C. Project Span

The project is limited to control door, garage, window status, smoke detection and temperature and luminosity control.

II. THEORY

Verilog is a hardware descriptive language. A Verilog module consists of various sub modules that can be combined to establish a complete system. This project is implemented by using verilog language. Xilinx Spartan 3E FPGA board is used for this project. All the sensors are interfaced with it.

A. *Hierarchy*

The next step is to create the hierarchy of the project. The diagram depicts the variety of conditions and states. The block diagram covers all the modules. It consists of main module which is divided into password and comfort module. Password module is meant to switch ON & OFF the security system. Password of 11 bit is created for the owner to enter inside the home. Security module transfers the information from the interfaced sensors to the connected inputs room_door, room_window, garage_door, fire. Door module monitors the state of the door persistently. If the owner finds open door due to the broken magnet, then the door alarm is set to one. Garage and window modules behave in a same way. Then it comes to the smoke module. If smoke detector detects the smoke then fire state will goes high which turn ON the fire alarm. Otherwise fire state remains low. When correct password is entered the security module gets activated which is further sub divided into door, window, garage and smoke modules. Comfort module is sub divided into temperature and luminosity controllers. Four types of alarms are there which include door alarm, window alarm, garage alarm and fire alarm.

Fig.1 shows the block diagram of entire home automation system.

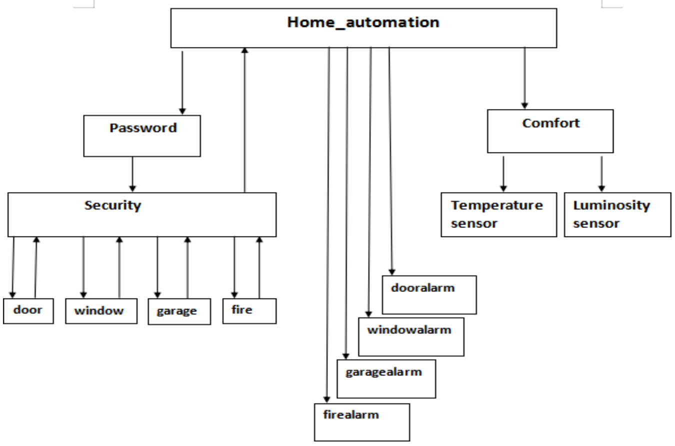

Fig .1 Home Automation System
[6]

B. *Flowchart*

Flowchart begin with Start state. An app is installed in the mobile phone which has android platform. Power is provided to HC05 and it is paired with the Bluetooth module. If it is not paired then it will go back to HC05 step. If BT is paired then FPGA and HC05 are connected through RS232. Application file is configured in FPGA. Instructions are given afterwards. FPGA receives the instruction and verifies it. If instruction is invalid pointer will go back to wait for valid instructions. If instruction is valid then project will work as per commands. Fig. 2 shows the flowchart of entire home automation system.

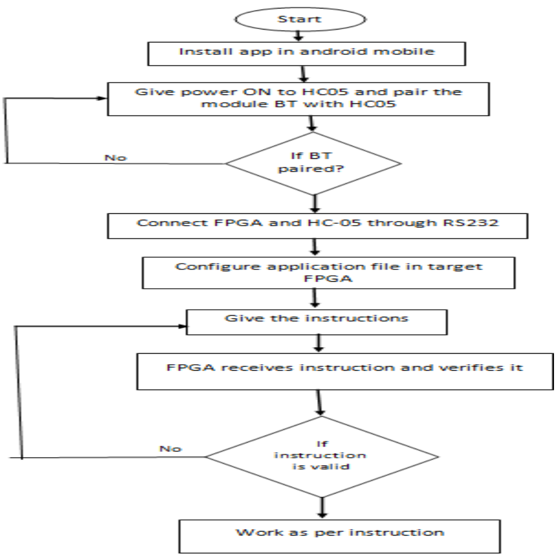

Fig 2 Flowchart of Home Automation Project

[8]

III. State Diagram for Comfort Module

The heart of comfort module is FSM i.e. Finite state machine. Here Moore machine is followed to accomplish the module. Moore machine is defined as FSM in which output is determined only from the current state while in case of Mealy output depends on both the current state and the value of its inputs. FSM consists of seven states in total. First one is start state or reset state. Here all the outputs of the module are zero. Then it comes to the temp heat state. In this state temperature of the room is checked by temperature sensor. If the temperature is below 15°C heater turn ON. If temperature is above 28°C then temp_cool state is executed and air conditioner starts. If the temperature is within 15 to 28° Celsius which is human comfort zone then temp_normal state gets executed. In all states light gets ON as the owner arrives inside the room. Then 3 more states are light_bright for bright light, light dim for dim light and light_normal for normal luminosity. Bright light gets on if luminosity is less than 200 Lux, dim light gets on if luminosity is more than 250 Lux. Light will neither get dim nor gets bright if luminosity is within 200 to 250 Lux range. Fig 3 shows FSM of the comfort module.

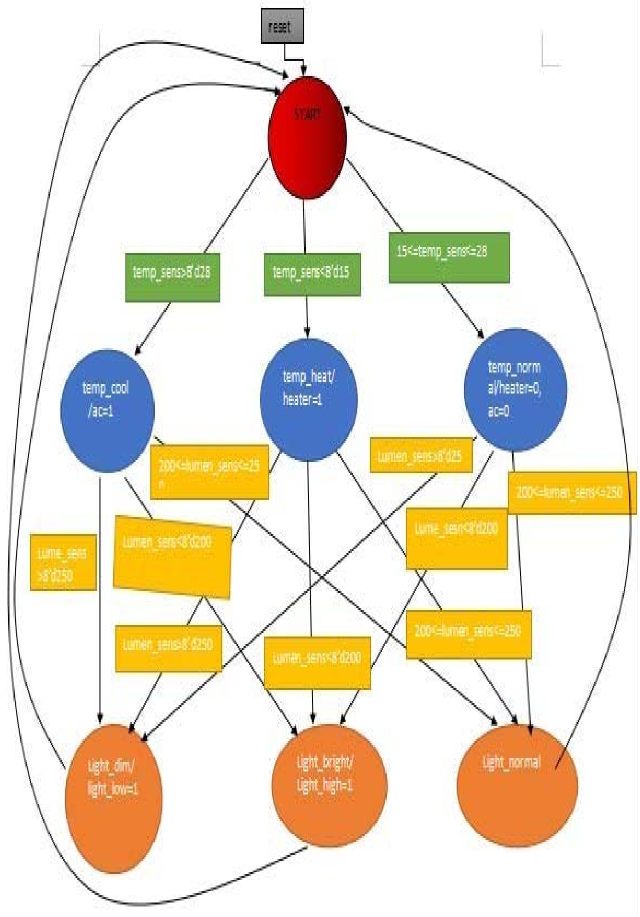

Fig.3 Comfort Module state diagram

[6]

IV. IMPLEMENTATION

This is an example of always block of garage module. On the positive edge of the clock when reset goes high, both garage state and garage alarm are low but as reset goes low, it will check garage door, if it is high then garage state is high and alarm will ring. If it is low then garage state will remains low and alarm will not ring. Other modules are similarly designed keeping in mind working of home automation modules.

```
always@(posedge clock)
begin
if(reset==1'b1)
begin
garage_state=1'b0;
garagealarm=1'b0;
end
else
begin
garage_state = garage_door ? 1'b1 : 1'b0; // go to burglary state if garage is on
garagealarm = (garage_state == 1'b1) ? (flag?1'b1:1'b0): 1'b0; // if burglary state, signal a burglary
end
end
```

V. RESULT

If the motion_sens that is the motion sensor is 1 then the comfort module gets activated otherwise all devices remains OFF. On the + ve transition of the clock lume_sen, temp_sen and motion_sen are checked. Various test cases have been taken and all the outputs are exhaustively checked and tested. Simulations results are depicted in the following figures. RTL view of comfort, security, top security modules are also shown. Waveform is obtained as desirable. Verilog Code is synthesizable. The results thus obtained will drive the whole home automation module.

A. *RTL Schematic*

Initially all the outputs are set to zero. considering a virtual house as role model the design is sketched. Interfacing of sensors is done thereafter. Every sensor is interfaced after that. Inputs include room_door, garage_door, room_window, smoke, alarms, states, light and temperature controllers. The RTL schematic is depicted in Fig.4 post synthesis on Xilinx platform which shows all inputs and outputs connected to a box. The input signals includes data_in [11:0] i.e. password to switch ON and OFF security. lumen_sens of 7 bits is signal provided by the light sensor & temp_sens of 7 bits is signal came from the temperature sensor, Clock for clock input, room_door is input from the magnetic door sensor, Smoke is the input to the smoke detector sensor, Garage_door is input for the garage laser sensor. Lastly motion_sens predict human's presence. room_window is input from the window magnetic sensor. In order to restart the FSM reset is used. The output signals includes door condition for opening and closing of door, fire status to find if there is smoke present in the atmosphere, window status for opening and closing window, garage output for opening and closing conditions of garage door, Alarm output for ringing the alarm on entering wrong password otherwise it does'nt ring, air conditioner gets ON when temp_sens > 28 degree celcius, the door alarm goes 1 when door is open, else it remains low, the fire alarm is 1 when smoke is appears, the flag is high when owner is inside home. RTL Schematic of comfort and top security is given in fig. 5. RTL Schematic of comfort, security and password is given in fig. 6. Simulation results are given in figure 7 and 8. Fig 7 represents inputs of home automation module and figure 8 represents its outputs.

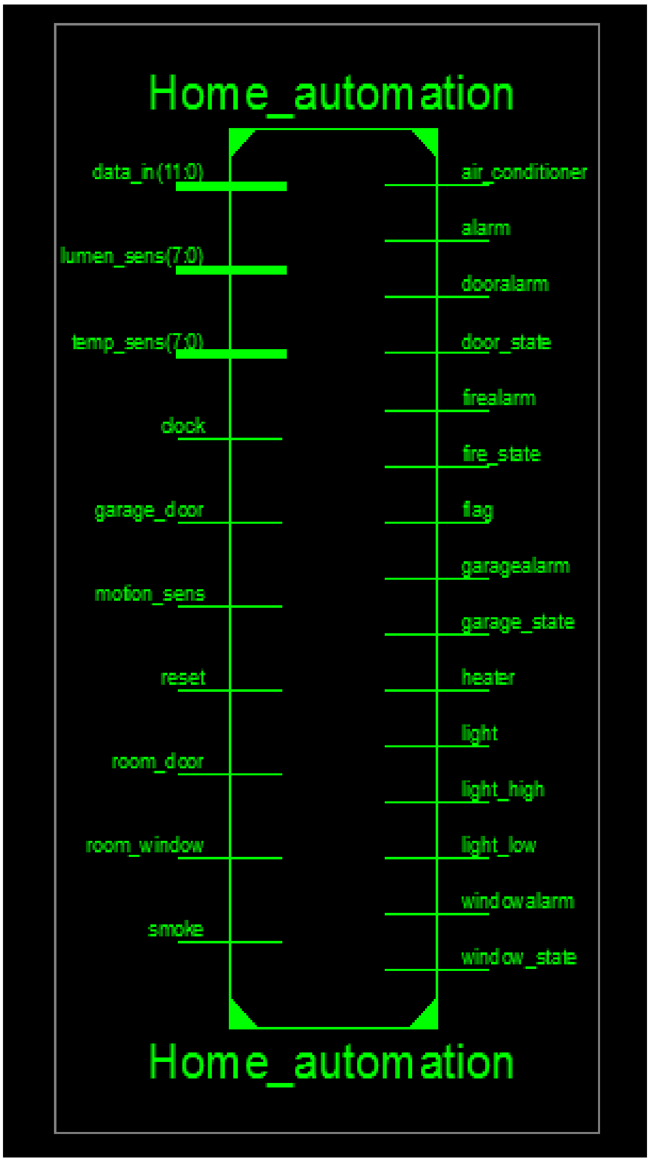

Fig. 4 Home Automation module

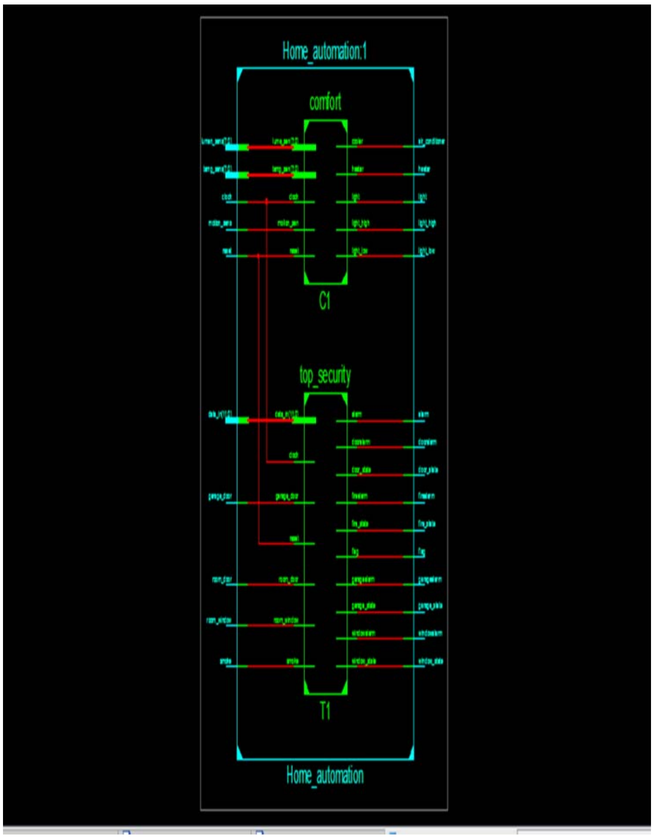

Fig.5 RTL schematic of comfort and top security

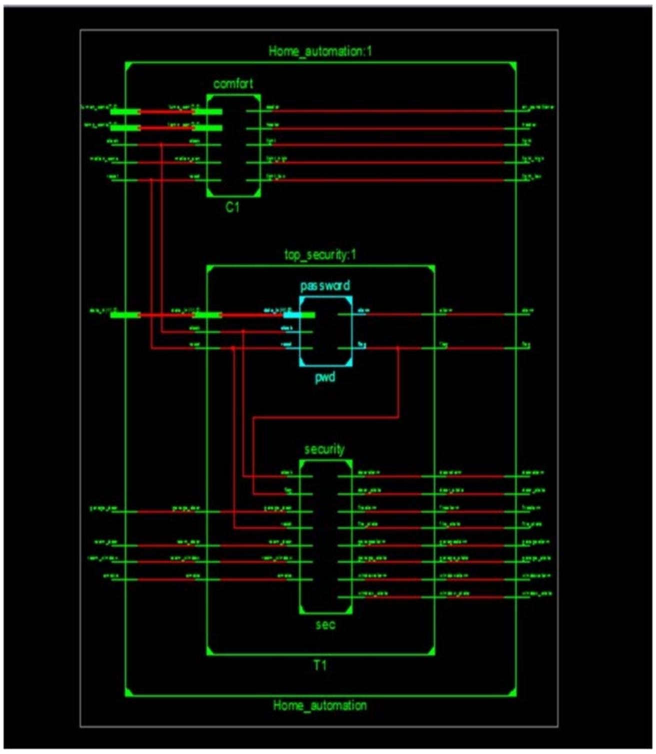

Fig.6 RTL schematic of comfort, password and security system

B. *Simulated Waveforms*

Lumen sensor reads 180 Lux and temperature sensor reads 15 degree Celsius. Since correct password is entered so owner enters the home. Since room window is high it means it is broken. These are the inputs of test case.

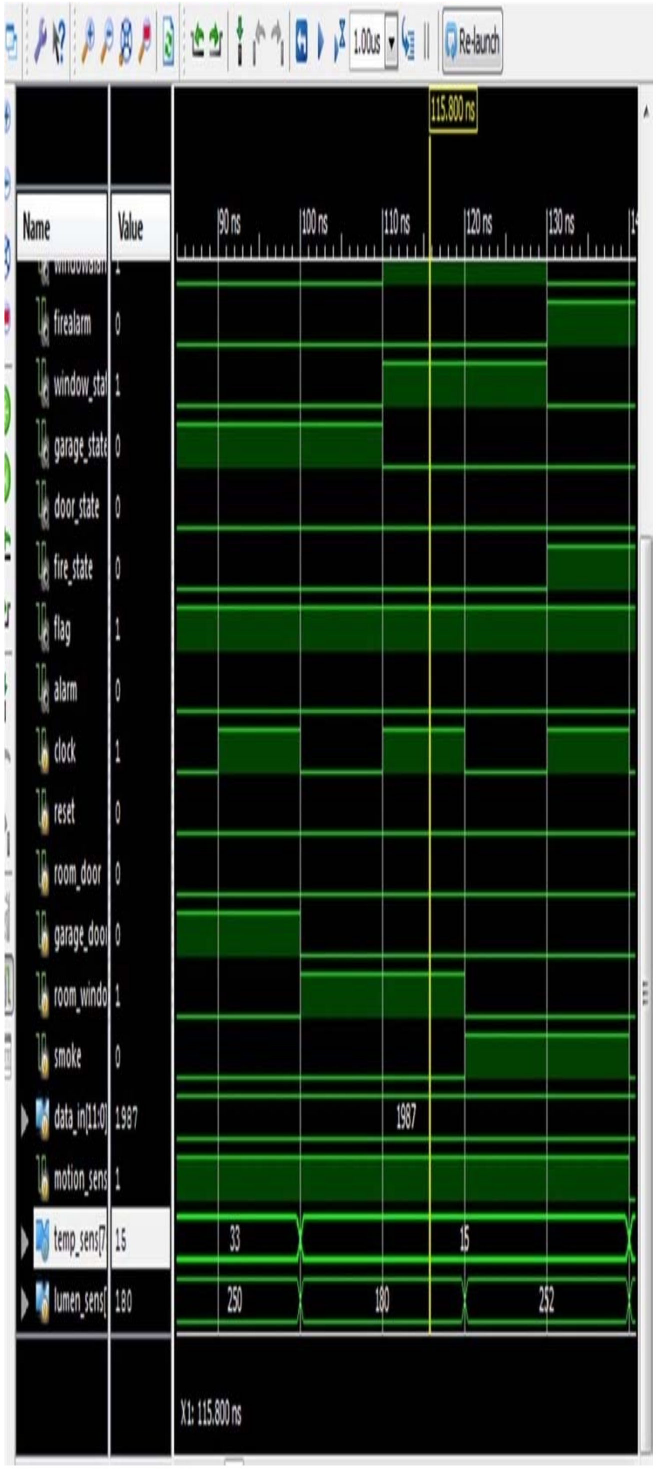

Fig.7 Xilinx output waveform after simulation

Flag will be 1 as owner enters into home. Only room window alarm will ring as its state is high. Light high will be on as luminosity is 180 Lux and neither heater nor AC will be ON as temperature is 15 degree Celsius.

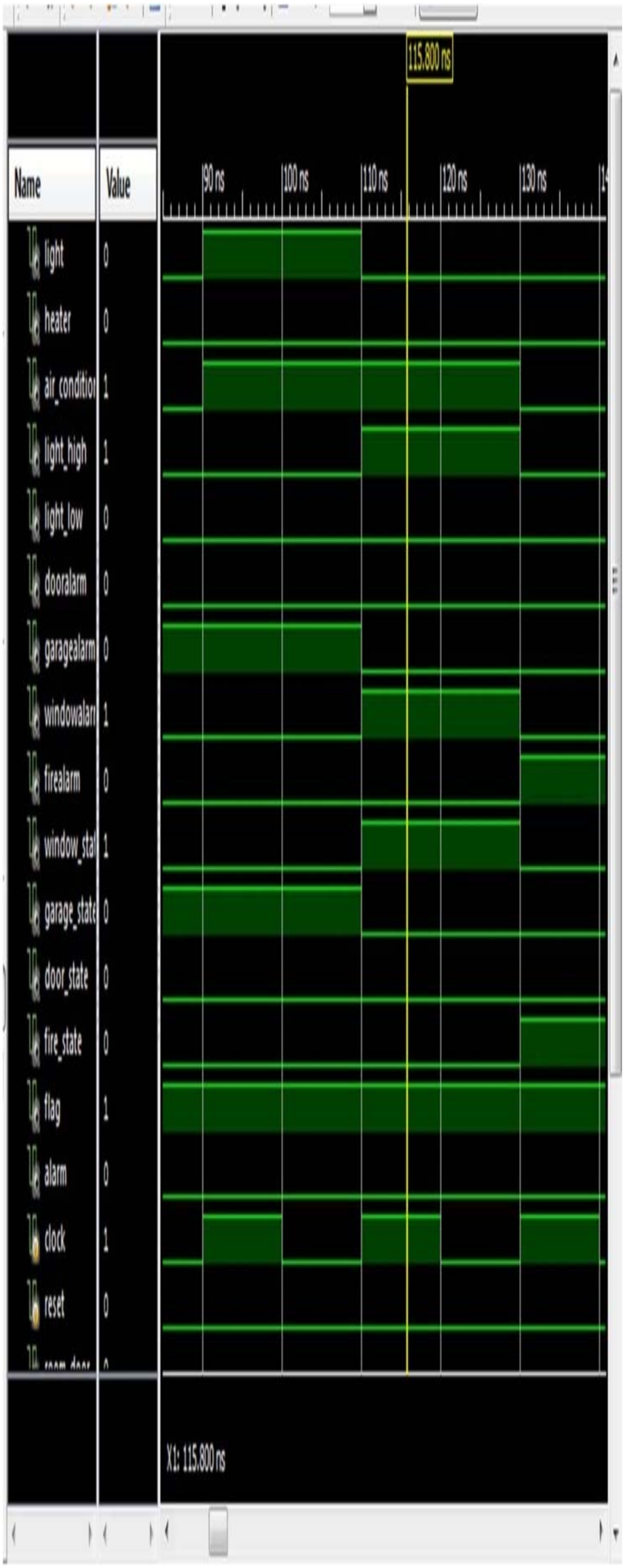

Fig.8 Xilinx output waveform after simulation

C. Device utilization summary

Table 1 Devce Utilization Summary Table

Slice logic utilization	**used**	**Available**	**utilization**
No. Used as flip flops	8		
No. Of slice LUTS	20	2400	1%
No. Of occupied slices	11	600	1%
No. Of bonded IOBs	50	102	49%

VI. CONCLUSION

Verilog provides RTL description which is advantageous for designer for debugging. RTL schematic can be modified by the designer and implemented by editing verilog code. The wave form is obtained as desired. All the test cases have been exhaustively checked. Thus it is inferred the smart home project is accomplished and it can be implemented without any flaw. Detailed utilization summary and reports are given in results.

VII. ACKNOWLEDGMENT

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